

Understanding Probability for Machine Learning – Events, Sample Space, and Basic Rules

In this lesson, we'll explore:

- What are events
- What is a sample space
- The basic rules of probability

By the end, you'll see how this simple math forms the foundation of almost every ML algorithm — from spam detection to self-driving cars.

1. What is Probability?

Probability is simply a **measure of uncertainty**.

It tells us *how likely* something is to happen.

Example:

- If you flip a fair coin, there's a **0.5 chance** (or 50%) it lands on heads.
- In ML, we might say there's a **0.9 probability** that an email is spam — meaning the model is 90% confident.

In short:

$$P(Event) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

2. Sample Space

The **sample space**, often denoted as **S**, is the set of *all possible outcomes* of an experiment.

Let's start simple:

- When you toss a coin:
 $S = \{\text{Heads, Tails}\}$
- When you roll a die:
 $S = \{1, 2, 3, 4, 5, 6\}$

In ML, we can think of the sample space as the **universe of all possible data points**.

Example:

- Suppose we're predicting if a student passes or fails based on study hours and attendance.
The sample space includes **all combinations** of study hours, attendance, and outcomes (pass/fail).

Key takeaway:

The sample space defines the world your probability model lives in. Just like your dataset defines the world your ML model learns from.

3. Events

An **event** is a subset of the sample space — something we care about happening.

Example:

- Rolling an even number $\rightarrow E = \{2, 4, 6\}$
- Getting heads $\rightarrow E = \{H\}$

In ML terms:

- Event = "The model predicts correctly"
- Event = "The data point belongs to class A"
- Event = "The user clicks on the ad"

Events can be:

- Simple event:** one outcome (like rolling a 3)
- Compound event:** multiple outcomes (like rolling an even number)
- Complementary event:** everything else (not rolling an even number)

We often use notation like:

$$P(E^c) = 1 - P(E)$$

That's just saying the probability of "not E" equals one minus the probability of "E".

4. Basic Probability Rules

Let's cover the three essential rules every ML student needs to know.

Rule 1: The Probability Range Rule

$$0 \leq P(E) \leq 1$$

A probability can never be negative or exceed 1.

If your ML model gives a probability greater than 1, something's definitely wrong with your math or scaling.

Rule 2: The Complement Rule

$$P(E) + P(E^c) = 1$$

Example:

If a model predicts that a user **will buy** a product with probability 0.7, then $P(\text{Not Buy}) = 1 - 0.7 = 0.3$.

This is exactly how classification models like **logistic regression** work. They compute one probability, and the rest follows automatically.

Rule 3: Addition Rule (for "OR" Events)

If A and B are **mutually exclusive** (can't happen together):

$$P(A \text{ or } B) = P(A) + P(B)$$

Example:

- Rolling a 2 or 3 on a die

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

If they **can** happen together, use the **general addition rule**:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

In ML:

- Suppose A = "User clicks on an ad"
 - B = "User buys the product"
- Some users might do both — that's $P(A \text{ and } B)$.

We subtract that overlap to avoid double-counting.

Rule 4: Multiplication Rule (for "AND" Events)

If A and B are **independent**:

$$P(A \text{ and } B) = P(A) \times P(B)$$

Example:

- Probability of getting two heads in a row
 $= 0.5 \times 0.5 = 0.25$

In ML, this independence assumption appears in models like **Naive Bayes**, where features are assumed to be independent given the class label.

That's why it's called *Naive* — but surprisingly, it works well in practice.

Bonus: Conditional Probability

This one's key for ML:

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

It reads: "Probability of A given B has occurred."

In ML:

- $P(\text{Spam} \mid \text{Has the word 'offer'})$
- The model learns these conditional probabilities from training data.

Conditional probability is the heart of **Bayes' Theorem**, which we'll explore later when we discuss **Naive Bayes classification**.

Section 5: Mini Real-World Use Cases

Let's connect this to real ML applications — so it doesn't just feel like math.

Example 1: Spam Filter

When you get an email, your ML model might check:

- Event A: It contains the word "Free"
- Event B: It contains "Offer"

It calculates probabilities like:

- $P(\text{Spam} \mid \text{Free})=0.8$
- $P(\text{Spam} \mid \text{Offer})=0.9$
- $P(\text{Spam} \mid \text{Free and Offer})=0.95$

That's the exact same math — just applied to real data!

Example 2: Disease Detection

In healthcare, models use probability to estimate the risk of disease.

For example:

- Event A: Test result is positive
- Event B: Patient actually has the disease

The model calculates how likely B is given A — that's where **Bayes' theorem** (coming next!) steps in.

6. Wrap Up

To recap:

- **Sample space** → all possible outcomes
- **Event** → the subset we care about
- **Probability** → measure of likelihood
- **Basic rules** → keep your probabilities logical and consistent

In ML, probability powers everything — from **classification**, **Bayesian inference**, **stochastic gradient descent**, to even **generative models**.

Next up, we'll use these concepts to understand **conditional probability and Bayes' theorem**, which directly connect to algorithms like **Naive Bayes** and **Probabilistic AI**.